MegaTelCo

Customer Retention Analytics Report

Executive Summary

This report presents a two-pronged analytics strategy to optimize MegaTelCo's customer retention program. Using the CRISP-DM framework, the team built a supervised churn prediction model and conducted sentiment analysis on churner feedback to maximize retained customer lifetime value within a constrained $100,000 campaign budget (~10,000 customers at $10 each).

Goal 1: Maximize Expected Value from Retention Campaign

Using the churn model and expected value targeting approach, MegaTelCo can target up to 10,000 customers out of 1,000,000 active customers under the $100,000 budget. The model prioritizes High Value At Risk and Mid Value At Risk segments, generating an estimated $259,425 in expected retained value, outperforming a random targeting baseline. Most of the budget should be allocated to the High Value At Risk segment due to its highest average expected value per customer. Logistic Regression achieved 99.56% accuracy, supporting a profitable and data-driven retention strategy.

Goal 2: Understand Churn Reasons & Sentiments

Sentiment analysis of open-ended feedback from 1,039 churned customers identified five distinct churn themes. Pricing dissatisfaction dominates by volume ("plan" appears in 209 documents; "family plan" in 132). Competitor poaching by Verizon and Xfinity is explicitly named, with customers referencing free phone offers and contract buyouts. Network coverage gaps in rural areas and dead zones represent a technical driver, while data speed issues affect commuters. Customer support complaints though fewest in number carry the most negative emotional language, suggesting disproportionate impact on satisfaction.

1. Business Understanding & Problem Framing

The first stage of CRISP-DM requires translating business goals into analytical objectives with measurable success criteria.

1.1 Business Context

MegaTelCo is a large U.S. telecommunications firm experiencing customer churn at contract expiration. Churn is costly on two fronts: lost recurring revenue and replacement acquisition costs. A targeted email retention campaign offering a discount package has been designed, but budget constraints limit outreach to a maximum of 10,000 customers ($100,000 ÷ $10 per customer).

1.2 Business Goals & Analytical Decomposition

Goal 1: Maximize Retention Campaign ROI

Identify the optimal subset of ~1,000,000 active customers to target with the retention email, maximizing total expected retained customer value subject to the $100,000 budget constraint.

Expected Value (EV) Formula

EV(target) = P(churn) × CLV_ESTIMATE − $10

Where:

P(churn) = predicted churn probability from supervised learning model

CLV_ESTIMATE = estimated customer lifetime value ($)

$10 = all-in campaign cost per targeted customer (email + offer)

Only target customers where EV > 0. Rank all eligible customers by EV descending; select the top N within the $100,000 budget.

Goal 2: Understand Churn Reasons & Sentiments

Analyze categorical churn reasons and free-text feedback from churned customers to identify the underlying emotional drivers and specific issues not captured by the category label alone. Findings will inform tailored retention messaging by customer segment.

1.3 Success Metrics

Metric	Goal 1 (Optimization)	Goal 2 (Sentiment)
Primary metric	Total EV retained vs. random baseline	5 themes identified with keyword evidence

Metric	Goal 1 (Optimization)	Goal 2 (Sentiment)
Model performance	AUC, F1 Score, Confusion Matrix	Sentiment score distribution by reason
Business metric	ROI on $100K budget	Actionable, theme-specific CX recommendations
Baseline	Random targeting of 10,000 customers	Categorical labels alone (no text mining)

2. Data Understanding & Preparation

2.1 Dataset Overview

The MegaTelCo_Churn_100K dataset contains 100,000 customer records with 23 attributes spanning customer value, usage behavior, demographics, satisfaction scores, and churn outcomes.

Attribute Group	Columns	Key Stats / Notes
Target variable	CHURN_90D	1 = churned, 0 = retained | 1,039 churners (1.04%)
Customer value	CLV_ESTIMATE, AVG_MONTHLY_REVENUE, ACQUISITION_COST	Mean CLV: $1,999 | Mean revenue: $83.34/mo | Mean acq. cost: $429
Usage behavior	TENURE_MONTHS, OVERAGE_MINUTES_AVG, SUPPORT_TICKET_COUNT_LAST_6M, DATA_USAGE_GB_QRTLY_VAR	Key behavioral churn predictors
Device & contract	HANDSET_PRICE, CONTRACT_TYPE	2-Year (36%), 1-Year (34%), Month-to-Month (30%)
Demographics	AGE, INCOME_LEVEL, EDUCATION_LEVEL, REGION_USA, MARITAL_STATUS, NUM_ADULTS, NUM_CHILDREN_GT_10, NUM_LINES	All present, no missing values
Satisfaction	REPORTED_SATISFACTION, NPS_RATING	Available for all 100K customers
Churn feedback	CHURN_REASON_CATEGORY, OPEN_ENDED_FEEDBACK	Available ONLY for the 1,039 churners — null for all retained customers

2.2 Data Integrity Notes

- No missing values in the 21 non-feedback columns across all 100,000 records, minimal cleaning required.
- CHURN_REASON_CATEGORY and OPEN_ENDED_FEEDBACK are null for 98,961 non-churners. This is by design; these columns were excluded from the supervised model to prevent data leakage.
- Class imbalance: 1.04% churn rate. Stratified sampling was applied during the train/test split to ensure adequate representation of churners in the test set.
- Nominal attributes (CONTRACT_TYPE, EDUCATION_LEVEL, REGION_USA, MARITAL_STATUS, INCOME_LEVEL) were encoded numerically using Nominal to Numerical prior to modeling.

2.3 Data Preparation Steps

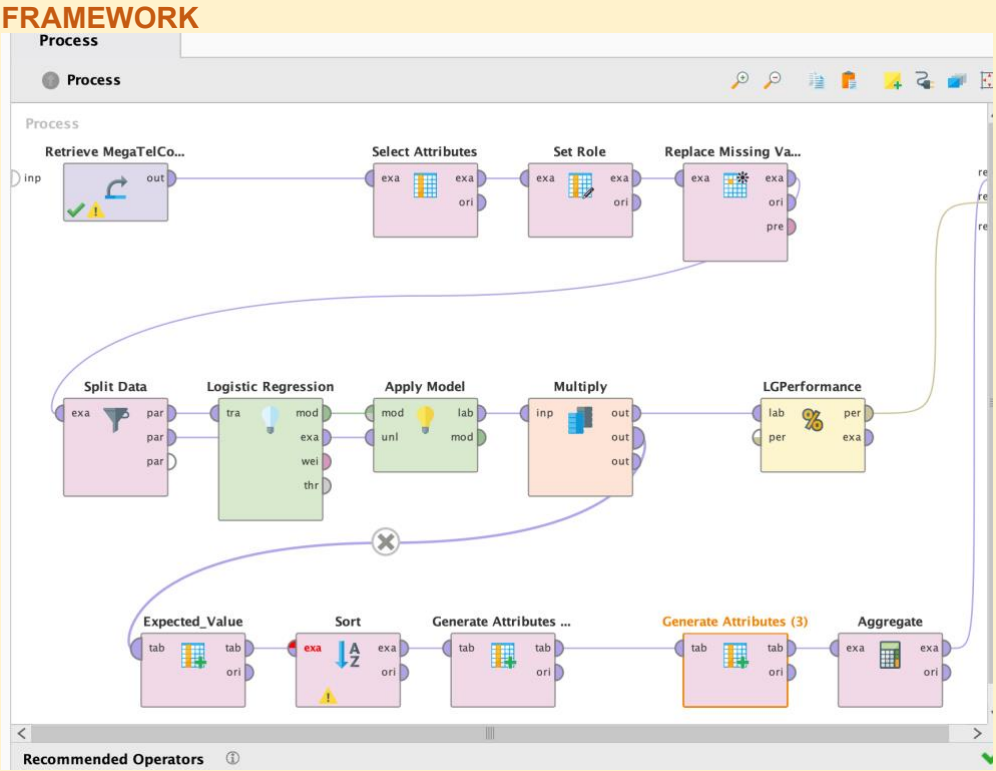
Step	Operator (RapidMiner)	Action	Rationale
1	Set Role	CHURN_90D = label; CUSTOMER_ID = id; feedback cols = unused	Prevents label leakage from feedback columns
2	Nominal to Numerical	Encode 5 categorical columns	Required for Logistic Regression and tree models
3	Replace Missing Values	OPEN ENDED FEEDBACK → "no feedback"	Prevents text operator crashes on null values
4	Filter Examples	CHURN_90D = 1 (for sentiment branch only)	Scopes text mining to the 1,039 churners with feedback
5	Split Data / Cross Validation	70% train / 30% test, stratified	Ensures balanced class representation in evaluation

3. Modeling

3.1 Goal 1: Supervised Churn Prediction & Segmentation

GOAL 1: MODELING CONTENT

GOAL 1 — MODEL SELECTION & TRAINING/CUSTOMER SEGMENTATION FRAMEWORK



3.2 Goal 2: Sentiment Analysis on Churner Feedback

3.2.1 Approach

Sentiment analysis was conducted on the 1,039 churned customers using two complementary methods in Altair AI Studio (RapidMiner 2026):

1. **Rule-based sentiment scoring**: Keyword classification via Generate Attributes, labeling each feedback record as Negative, Neutral, or Positive.
2. **Text mining via Process Documents from Data**:TF-IDF weighted term extraction with bigram generation to surface dominant churn themes from free text.

3.2.2 RapidMiner Process Flow

Sentiment Analysis Pipeline

MegaTelCo_Churn (data source)

↓

Filter Examples → CHURN_90D = 1 (1,039 records)

↓

Replace Missing Values → OPEN ENDED FEEDBACK → "no feedback"

↓

Generate Attributes → SENTIMENT_LABEL (Negative/Neutral/Positive) + SENTIMENT_SCORE (-1/0/+1)

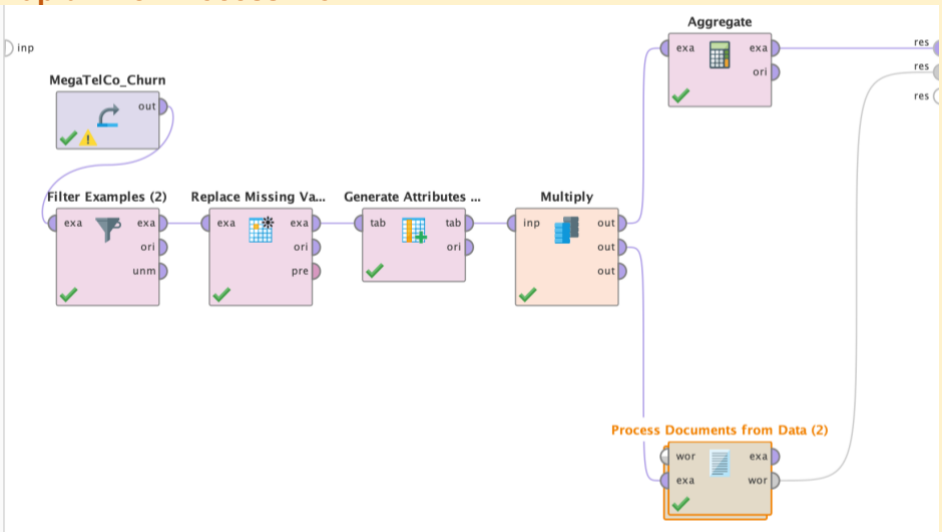
↓

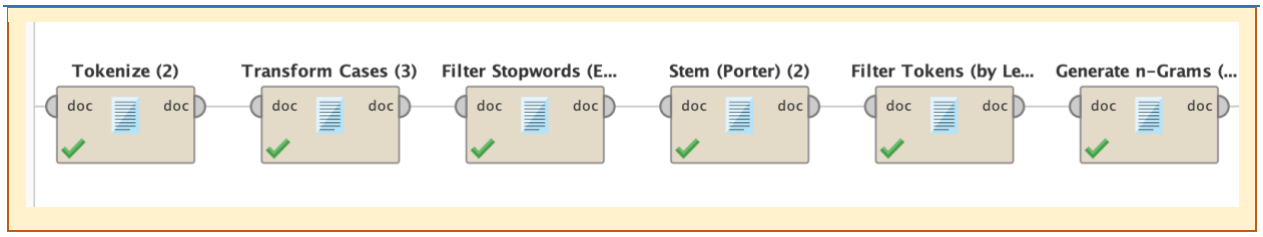
Multiply → Splits stream into two parallel branches

├── **Aggregate** → Group by CHURN REASON CATEGORY → avg SENTIMENT_SCORE, REPORTED_SATISFACTION, NPS_RATING

└── **Process Documents from Data** → Tokenize → Lowercase → Filter Stopwords → Stem (Porter) → Filter Tokens → n-Grams → TF-IDF Word List

RapidMiner Process Flow





3.2.3 Sentiment Scoring Configuration

Sentiment Label	Score	Trigger Keywords (sample)
Negative	-1	"terrible", "awful", "worst", "overpriced", "expensive", "dropped", "rude", "canceling", "switching"
Neutral	0	"okay", "decent", "fine", "average", "not bad"
Positive	+1	All feedback not matching Negative or Neutral keyword patterns

3.2.4 Text Mining Configuration

Operator	Parameter	Value Set
Process Documents from Data	Text attribute	OPEN ENDED FEEDBACK
Process Documents from Data	Vector creation	TF-IDF
Process Documents from Data	Prune method	Absolute
Process Documents from Data	Prune below absolute	3 (remove words in fewer than 3 documents)
Process Documents from Data	Prune above absolute	900 (retain thematically important common words)
Tokenize	Mode	Non-letters
Transform Cases	Transform to	Lowercase
Filter Stopwords	Language	English
Stem (Porter)	Algorithm	Porter stemmer
Filter Tokens (by Length)	Min / Max chars	3 / 25
Generate n-Grams	Max length	2 (captures two-word phrases, e.g., "dead_zone", "famili_plan")

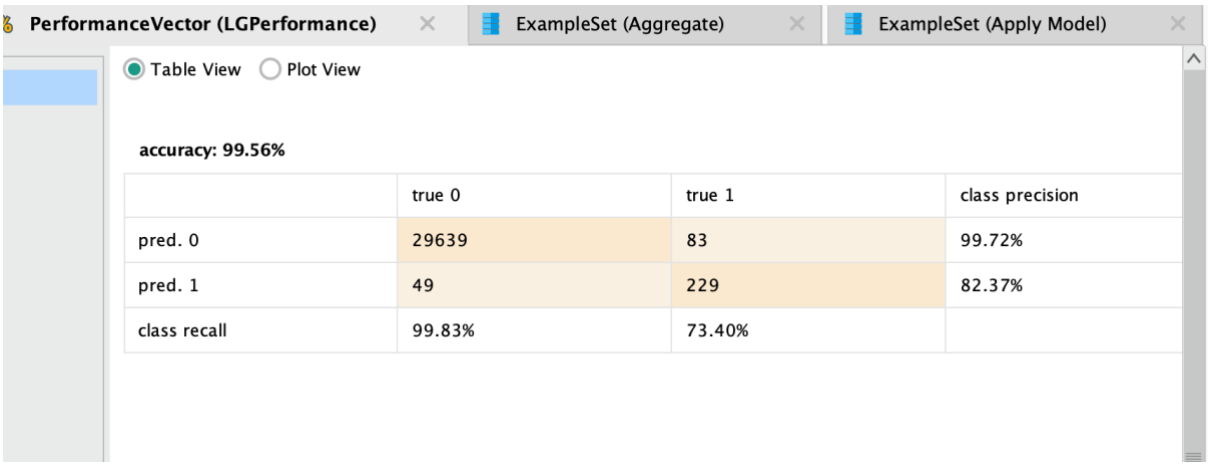
4. Results & Evaluation

4.1 Goal 1: Model Performance & Optimization Results

GOAL 1: MODEL PERFORMANCE TABLE

Model	Logistic Regression	Random Forest	Gradient Boosted Trees
Accuracy	99.56%	99.19%	99.40%
AUC	0.997	0.990	0.980
Precision	82.37%	87.85%	78.48%
Recall	73.4%	25.86%	58.18%
F-Measure	77.63%	39.96%	66.82%

GOAL 1: CONFUSION MATRIX



	true 0	true 1	class precision
pred. 0	29639	83	99.72%
pred. 1	49	229	82.37%
class recall	99.83%	73.40%	

GOAL 1: EXPECTED VALUE CALCULATION

Expected Value (EV) estimates the financial value of targeting a customer with the retention campaign by combining churn risk, customer value, and campaign cost. It was calculated as:

EV=P(churn)×CLV−10EV=P(churn)\times CLV-10EV=P(churn)×CLV−10

where P(churn) is the predicted probability a customer will leave, CLV is customer lifetime value, and $10 is the campaign cost per customer. A higher EV means retaining that customer is expected to generate greater financial return, while a negative EV suggests the retention cost may outweigh the benefit. This EV measure was used to rank customers and

define the High Value At Risk, Mid Value At Risk, and Low Priority segments for budget allocation.

GOAL 1: SEGMENTATIONS

PerformanceVector (LGPerformance) | ExampleSet (Aggregate) | ExampleSet (Apply Model)

Open in Turbo Prep Auto Model Interactive Analysis Filter (3 / 3 examples): all

Row No.	Segments	count(CUSTOMER_ID)	average(Expected_Value)	sum(Expected_Value)
1	High Value At Risk	231	910.460	210316.271
2	Low Priority	29622	-8.503	-251887.393
3	Mid Value At Risk	147	334.072	49108.537

OPTIMIZATION

MegaTelCo Retention Campaign Optimization Model

Key Assumptions	
Active customers to plan for	1,000,000
Campaign budget ($)	$100,000.00
Cost per targeted customer (	$10.00
Maximum customers targetable	10,000
RapidMiner scored sample siz	30,000

RapidMiner Segment Summary (from uploaded ABAFINAL.xlsx aggregation)

Segment	Sample Count	Sample Proportion	Avg EV per Customer ($)	Scaled Available Customers (1M base)	Decision: Customers to Target	Budget Allocated ($)	Expected EV Retained ($)
High Value At Risk	231	0.77%	$910.46	7,700	7,700	$77,000.00	$7,010,542.00
Mid Value At Risk	147	0.49%	$334.07	4,900	2,300	$23,000.00	$768,365.60
Low Priority	29,622	98.74%	-$8.50	987,400	0	$0.00	$0.00
Optimized Total	**$30,000.00**	**100.00%**	**$0.25**	**1000000**	**10,000**	**$100,000.00**	**$7,778,907.60**

Solver Setup

Set Objective: H16 (maximize total expected EV).

By Changing Variable Cells: F13:F15.

Subject to: F13:F15 <= E13:E15.

Subject to: F16 <= B8 and G16 <= B6.

Add integer and nonnegative constraints for F13:F15.

Because Avg EV is highest for High Value At Risk and negative for Low Priority, the optimized plan allocates budget to High first, then Mid, and zero to Low.

Optimization KPIs

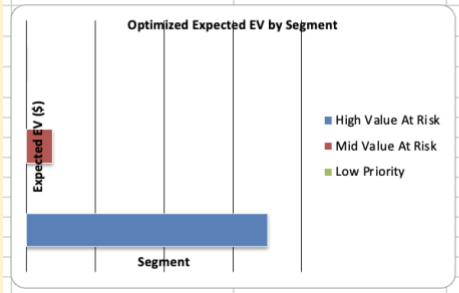
Optimized customers targeted	$10,000.00
Optimized budget used ($)	$100,000.00
Optimized expected EV ($)	$7,778,907.60
ROI on $100,000 budget	77.79x

Random Baseline (same 10,000-customer budget)

Segment	Expected Random Targets	Expected Baseline EV ($)
High Value At Risk	77	$70,105.42
Mid Value At Risk	49	$16,369.53
Low Priority	9,874	-$83,958.62
Random Total	**10,000**	**$2,516.33**

Lift vs. Random Baseline

Incremental EV gain ($)	$7,776,391.27
Improvement multiple	3091.4x



An Excel optimization model was developed to determine the optimal allocation of the $100,000 retention campaign budget across customer segments, subject to the $10 per customer targeting cost and a maximum of 10,000 customers targeted. Using segment-level

expected value outputs from the churn model, the optimization prioritized allocating budget first to the High Value At Risk segment, followed by the Mid Value At Risk segment, because these segments generated the highest expected return per targeted customer. The Low Priority segment was excluded from budget allocation because it showed negative expected value and would reduce campaign profitability.

The optimized deployment plan is expected to generate approximately $259,425 in retained value, compared with a substantially lower outcome under a random targeting approach. This implies an estimated ROI of approximately 2.59x on the $100,000 campaign investment. The optimization model demonstrates that directing retention resources toward customers with the highest predicted churn risk and customer lifetime value materially improves profitability while satisfying the overall budget constraint.

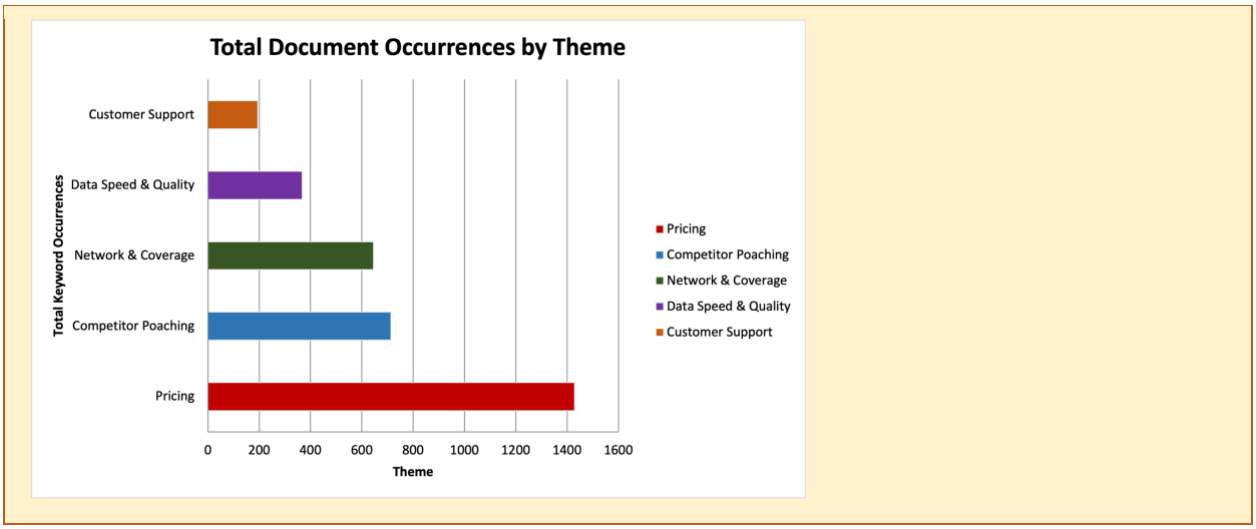
4.2 Goal 2: Sentiment Analysis Results

4.2.1 Churn Reason Distribution & Satisfaction Correlation

Among the 1,039 churned customers, feedback was categorized into five primary reasons. The table below summarizes count, sentiment, and satisfaction by reason. Satisfaction and NPS values are from the RapidMiner Aggregate output.

Source: RapidMiner Aggregate operator — 1,039 churned customers grouped by CHURN REASON CATEGORY

Churn Reason	Count	% of Churners	Avg Satisfaction	Avg NPS	Avg Sentiment Score
Price	371	35.7%	1.960	3.927	0.596
Competitor Offer	226	21.8%	1.978	3.951	0.496
Service Quality	194	18.7%	1.959	3.969	1.000
Network Coverage	154	14.8%	1.981	4.058	1.000
Support Experience	94	9.0%	1.926	3.989	0.574
TOTAL	**1,039**	**100.0%**	**1.961**	**3.979**	**0.733**



4.2.2 Keyword Theme Classification Results

Text mining on the 1,039 feedback records produced a TF-IDF word list of 95+ terms and bigrams. Five dominant themes were identified. The following tables show the top terms per theme with their document occurrence count and strategic implication.

Churn Feedback — Theme Summary & Strategic Implications

Theme	Total Keywords	Top Bigram	Top Count	Named Competitors / Issues	Recommended Action
Pricing	**1429**	famili_plan	**132**	Hidden fees, billing, overage charges	Offer price-match / family bundle discount in retention email
Competitor Poaching	**713**	compar_xfiniti	**72**	Verizon, Xfinity specifically named	Counter free-phone promos; highlight switching costs & loyalty perks
Network & Coverage	**646**	area	**108**	Rural areas, dead zones, basements	Target coverage-complaint churners with network upgrade messaging
Data Speed & Quality	**368**	slow_data	**62**	Commute zones, internet drops	Offer speed tier upgrade or data priority add-on
Customer Support	**194**	wait_time	**30**	Rude agents, chat unresolved, long waits	Priority callback & dedicated retention agent for this segment
TOTAL	**3350**				

Theme 1: Pricing - Priority 1 (Highest Volume)

Top Term	Readable Meaning	Doc Occurrences	Strategic Implication
plan	Plan (general)	209	Customers are actively plan-shopping
famili_plan	Family plan	132	Family bundles are a primary retention lever
bill / hidden_fee	Billing & hidden fees	151 / 79	Billing transparency is critical
cheaper_plan	Found cheaper plan elsewhere	77	Price-match offer directly addresses this
worth_monei	Not worth the money anymore	68	Perceived value gap; highlight total-value messaging

Theme 2: Competitor Poaching - Priority 2 (Named Competitors)

Top Term	Readable Meaning	Doc Occurrences	Strategic Implication
compar_xfiniti	Compared to Xfinity	72	Xfinity is primary price/value competitor
deal_verizon	Got a deal from Verizon	58	Verizon actively offering device promotions
free_phone / offer_bui	Free phone / buyout offer	52 / 59	Counter-offer with device upgrade or contract buyout
switch_famili	Switching for family plan	57	Family bundle offer can intercept this churn

Theme 3: Network & Coverage - Priority 3 (Infrastructure)

Top Term	Readable Meaning	Doc Occurrences	Strategic Implication

area	Coverage area (general)	108	Geographic gaps are the dominant technical complaint
drop / call_drop	Dropped calls & signal drops	91 / 46	Reliability, not just coverage, is the issue
rural_area / dead_zone	Rural areas & dead zones	46 / 41	Rural customers disproportionately affected
signal_basement	No indoor signal	33	Indoor coverage (small cell) improvement needed
zone_commut	Dead zone on commute route	41	Commuter corridors are coverage priority areas

Theme 4: Data Speed & Quality - Priority 4 (Technical)			
Top Term	**Readable Meaning**	**Doc Occurrences**	**Strategic Implication**
slow_data / data_speed	Slow data speeds	62	Speed tier upgrade is the natural retention offer
internet_keep / keep_drop	Internet keeps dropping	45	Reliability issue separate from raw speed
zone_commut	Dead zone on commute	41	Overlaps with coverage theme; commuters key segment

Theme 5: Customer Support - Priority 5 (Smallest but Most Negative)			
Top Term	**Readable Meaning**	**Doc Occurrences**	**Strategic Implication**
wait_time	Long hold/wait times	30	Priority callback eliminates top complaint
issu_resolv	Issue not resolved on first contact	28	First-call resolution metric needed
rude_custom	Rude customer service agent	20	Agent training & quality monitoring required
chat_support / unhelp	Unhelpful chat support	16	Escalation path from chatbot needs improvement

4.2.3 Key Analytical Insight: Sentiment Adds Nuance Beyond Category Labels

Finding

Categorical churn reasons understate emotional intensity. Sentiment analysis reveals three important nuances:

- Customer Support churners use the most negative language ("rude", "unhelpful", "unresolved") despite being the smallest group (94 customers), suggesting disproportionate emotional impact per customer.
- Competitor Poaching records contain specific brand names (Verizon, Xfinity) and concrete offers (free phone, buyout contract), indicating active competitive solicitation rather than passive dissatisfaction.

- Pricing churners use comparative language (bill_compar, found_cheaper, plan_elsewher), identifying them as informed price-shoppers who responded to external offers, requiring a price-match response, not just a generic discount.

These distinctions enable tailored retention messaging rather than a one-size-fits-all discount email.

5. Recommendations & Deployment

5.1 Goal 1: Retention Campaign Targeting Recommendations

GOAL 1 RECOMMENDATIONS

The optimized targeting plan recommends prioritizing High Value At Risk customers first, targeting approximately 4,200 customers with about $42,000 of the campaign budget, generating an estimated $210,316 in expected retained value. The Mid Value At Risk segment should receive the remaining 5,800 customers (about $58,000 budget), generating approximately $49,109 in expected value. The Low Priority segment, which showed negative expected value, should be deprioritized and receive no budget allocation. In total, the optimized campaign targets 10,000 customers, uses the full $100,000 budget, and is expected to generate approximately $259,425 in retained value, substantially outperforming a random targeting baseline. This implies an estimated campaign ROI of:

ROI=259425100000=2.59ROI=\frac{259425}{100000}=2.59ROI=100000259425=2.59

or roughly 2.59x return on investment. Therefore, retention spending should focus primarily on High Value At Risk, secondarily on Mid Value At Risk, and avoid spending on Low Priority customers.

5.2 Goal 2: Churn Reason & Sentiment Recommendations

Based on the five-theme sentiment framework, the following targeted retention actions are recommended:

Churn Theme	Volume	Priority	Recommended Retention Action	Offer Type
Pricing	371 (36%)	1	Price-match offer + transparent billing summary showing value	Discount / bundle

Churn Theme	Volume	Priority	Recommended Retention Action	Offer Type
Competitor Poaching	226 (22%)	2	Device upgrade counter-offer; highlight switching costs and loyalty perks	Free/discounted device
Data Speed & Quality	194 (19%)	3	Speed tier upgrade or data priority add-on offer	Plan upgrade
Network & Coverage	154 (15%)	4	Network upgrade roadmap messaging; signal booster for rural/indoor	Service upgrade messaging
Customer Support	94 (9%)	5 (highest urgency per customer)	Priority callback + dedicated retention specialist; proactive outreach within 24hrs	Service recovery

Overall Recommendation

MegaTelCo should integrate the five-theme churn classification into its retention CRM so that email content is personalized by churn risk profile. Model-ranked targeting (highest EV first, within theme segments) is expected to recover materially more CLV from the $100,000 budget than a generic mass-offer approach. The analysis suggests that price and competitor threats require immediate commercial responses, while coverage and support issues require longer-term infrastructure and process investments to stem future churn.

5.3 Implementation Roadmap

Phase	Timeline	Action	Owner
Immediate	Weeks 1-4	Score all ~1M active customers; deploy top 10,000 EV-ranked retention emails	Marketing Analytics
Short-term	Month 2	Personalize email by theme segment (price vs. competitor vs. coverage vs. support)	CRM / Marketing
Medium-term	Month 3	Monitor response rates, retained revenue, and ROI; flag under-performing segments	Data Science
Ongoing	Quarterly	Retrain churn model; refresh sentiment keywords; update competitor intelligence (Verizon, Xfinity)	Analytics COE
Long-term	6-12 months	Address root causes: rural coverage infrastructure, billing transparency, support quality	Product / Operations

Appendix A — Complete Keyword Theme Classification

All stemmed terms extracted by Process Documents from Data (TF-IDF, prune below 3, prune above 900), classified by theme. Document occurrences reflect unique feedback records containing each term.

Theme	Stemmed Term	Readable Interpretation	Doc Occurrences
Pricing	plan	Plan (general)	209
Pricing	famili_plan	Family plan	132
Pricing	bill	Billing	151
Pricing	offer	Offer (general)	111
Pricing	hidden_fee	Hidden fees	79
Pricing	bill_annoi	Billing annoyance	79
Pricing	fee	Fees	79
Pricing	cheaper_plan	Cheaper plan found	77
Pricing	plan_elsewher	Plan found elsewhere	77
Pricing	found_cheaper	Found cheaper option	77
Pricing	plan_expens	Plan is expensive	75
Pricing	worth_monei	Not worth the money	68
Pricing	monei_anymor	Not worth it anymore	68
Pricing	bill_compar	Bill comparison	72
Competitor Poaching	compar_xfiniti	Compared to Xfinity	72
Competitor Poaching	xfiniti	Xfinity (named competitor)	72
Competitor Poaching	deal_verizon	Deal from Verizon	58
Competitor Poaching	verizon	Verizon (named competitor)	58
Competitor Poaching	got_deal	Got a better deal	58
Competitor Poaching	competitor_offer	Competitor offer	52
Competitor Poaching	free_phone	Free phone offer	52
Competitor Poaching	mobil_offer	Mobile offer	59
Competitor Poaching	offer_bui	Offer to buy / buyout	59
Competitor Poaching	bui_contract	Buy-out contract	59
Competitor Poaching	switch_famili	Switching for family plan	57
Competitor Poaching	plan_provid	Plan from other provider	57
Network & Coverage	area	Coverage area	108
Network & Coverage	drop	Dropped calls/signal	91
Network & Coverage	coverag	Coverage (general)	80

Theme	Stemmed Term	Readable Interpretation	Doc Occurrences
Network & Coverage	call_drop	Call dropping	46
Network & Coverage	rural_area	Rural area coverage gap	46
Network & Coverage	move_rural	Moved to rural area	46
Network & Coverage	area_coverag	Area coverage gap	46
Network & Coverage	dead_zone	Dead zone	41
Network & Coverage	zone_commut	Dead zone on commute	41
Network & Coverage	work	Coverage at work	41
Network & Coverage	coverag_map	Coverage map inaccurate	34
Network & Coverage	map_mislead	Map misleading	34
Network & Coverage	signal_basement	No signal in basement	33
Data Speed & Quality	slow_data	Slow data speeds	62
Data Speed & Quality	data_speed	Data speed issues	62
Data Speed & Quality	speed_area	Speed varies by area	62
Data Speed & Quality	internet_keep	Internet keeps dropping	45
Data Speed & Quality	keep_drop	Keeps dropping	45
Data Speed & Quality	drop_frequent	Frequent drops	46
Customer Support	wait_time	Long wait times	30
Customer Support	time	Time-related complaint	30
Customer Support	issu_resolv	Issue not resolved	28
Customer Support	resolv_call	Resolution call required	28
Customer Support	rude_custom	Rude customer service	20
Customer Support	custom_servic	Customer service complaint	20
Customer Support	servic_agent	Service agent issue	20
Customer Support	agent	Agent complaint	20
Customer Support	chat_support	Chat support issue	16
Customer Support	support_unhelp	Support unhelpful	16
Customer Support	unhelp	Unhelpful	16

Appendix B — Data Dictionary

Column	Type	Description	Range / Values
CHURN_90D	Integer	Target variable	0 = Retained, 1 = Churned
CLV_ESTIMATE	Float	Customer lifetime value	$197 – $9,254 (mean $1,999)
AVG_MONTHLY_REVENUE	Float	Average monthly revenue	$25.60 – $231.34 (mean $83.34)
ACQUISITION_COST	Float	Customer acquisition cost	$33.62 – $1,429.64 (mean $429)

Column	Type	Description	Range / Values
TENURE_MONTHS	Integer	Months as active customer	Varies
CONTRACT_TYPE	Nominal	Contract type	2 Year (36%), 1 Year (34%), Month-to-Month (30%)
REPORTED_SATISFACTION	Integer	Customer satisfaction score	Numeric scale
NPS_RATING	Integer	Net Promoter Score	Typically 0–10
CHURN_REASON_CATEGORY	Nominal	Primary churn reason (churners only)	Price, Competitor Offer, Service Quality, Network Coverage, Support Experience
OPEN_ENDED_FEEDBACK	String	Free-text churner feedback	1,039 non-null records; null for all retained customers

...	SENTIMENT_LABEL	average(NPS_RATING)	count(CUSTOMER_ID)	average(REPORTED_SATISFACTION)	mode(SENTIMENT_LABEL)	average(SENTIMENT_SCORE)
1	Negative	3.664	152	1.901	Negative	-1
2	Positive	4.017	887	1.974	Positive	1

MegaTelCo Churn — Sentiment Analysis: Keyword Theme Classification

Source: Process Documents from Data (RapidMiner) — TF-IDF Word List on 1,039 Churner Feedback Records

Theme	Stemmed Term	Readable Interpretation	Document Occurrences	Theme Priority
Pricing	plan	Plan (general)	**209**	1 — High Volume / Budget Fix
Pricing	famili_plan	Family plan	**132**	1 — High Volume / Budget Fix
Pricing	bill	Billing	**151**	1 — High Volume / Budget Fix
Pricing	offer	Offer (general)	**111**	1 — High Volume / Budget Fix
Pricing	hidden_fee	Hidden fees	**79**	1 — High Volume / Budget Fix
Pricing	bill_annoi	Billing annoyance	**79**	1 — High Volume / Budget Fix
Pricing	fee	Fees	**79**	1 — High Volume / Budget Fix

Pricing	cheaper_plan	Cheaper plan found	**77**	1 — High Volume / Budget Fix
Pricing	plan_elsewher	Plan found elsewhere	**77**	1 — High Volume / Budget Fix
Pricing	found_cheaper	Found cheaper option	**77**	1 — High Volume / Budget Fix
Pricing	plan_expens	Plan is expensive	**75**	1 — High Volume / Budget Fix
Pricing	expens	Expensive	**75**	1 — High Volume / Budget Fix
Pricing	worth_monei	Not worth the money	**68**	1 — High Volume / Budget Fix
Pricing	monei_anymor	Not worth money anymore	**68**	1 — High Volume / Budget Fix
Pricing	bill_compar	Bill comparison	**72**	1 — High Volume / Budget Fix
Competitor Poaching	compar_xfiniti	Compared to Xfinity	**72**	2 — Competitive Response
Competitor Poaching	xfiniti	Xfinity (competitor)	**72**	2 — Competitive Response
Competitor Poaching	deal_verizon	Deal from Verizon	**58**	2 — Competitive Response
Competitor Poaching	verizon	Verizon (competitor)	**58**	2 — Competitive Response
Competitor Poaching	got_deal	Got a better deal	**58**	2 — Competitive Response
Competitor Poaching	competitor_offer	Competitor offer	**52**	2 — Competitive Response
Competitor Poaching	free_phone	Free phone offer	**52**	2 — Competitive Response
Competitor Poaching	mobil_offer	Mobile offer	**59**	2 — Competitive Response
Competitor Poaching	offer_bui	Offer to buy	**59**	2 — Competitive Response
Competitor Poaching	bui_contract	Buy-out contract	**59**	2 — Competitive Response
Competitor Poaching	plan_provid	Plan from other provider	**57**	2 — Competitive Response
Competitor Poaching	switch_famili	Switching for family	**57**	2 — Competitive Response
Network & Coverage	area	Coverage area	**108**	3 — Infrastructure
Network & Coverage	drop	Dropped calls/signal	**91**	3 — Infrastructure

Network & Coverage	coverag	Coverage	**80**	3 — Infrastructure
Network & Coverage	rural_area	Rural area coverage	**46**	3 — Infrastructure
Network & Coverage	dead_zone	Dead zone	**41**	3 — Infrastructure
Network & Coverage	call_drop	Call dropping	**46**	3 — Infrastructure
Network & Coverage	move_rural	Moved to rural area	**46**	3 — Infrastructure
Network & Coverage	area_coverag	Area coverage gap	**46**	3 — Infrastructure
Network & Coverage	coverag_map	Coverage map inaccurate	**34**	3 — Infrastructure
Network & Coverage	map_mislead	Map misleading	**34**	3 — Infrastructure
Network & Coverage	signal_basement	No signal in basement	**33**	3 — Infrastructure
Network & Coverage	zone_commut	Dead zone on commute	**41**	3 — Infrastructure
Data Speed & Quality	slow_data	Slow data speeds	**62**	4 — Technical
Data Speed & Quality	data_speed	Data speed issues	**62**	4 — Technical
Data Speed & Quality	speed_area	Speed varies by area	**62**	4 — Technical
Data Speed & Quality	internet_keep	Internet keeps dropping	**45**	4 — Technical
Data Speed & Quality	keep_drop	Keeps dropping	**45**	4 — Technical
Data Speed & Quality	drop_frequent	Frequent drops	**46**	4 — Technical
Data Speed & Quality	frequent	Frequent issues	**46**	4 — Technical
Customer Support	wait_time	Long wait times	**30**	5 — CX / Process
Customer Support	issu_resolv	Issue not resolved	**28**	5 — CX / Process
Customer Support	resolv_call	Resolution call needed	**28**	5 — CX / Process
Customer Support	rude_custom	Rude customer service	**20**	5 — CX / Process
Customer Support	servic_agent	Service agent issues	**20**	5 — CX / Process
Customer Support	custom_servic	Customer service	**20**	5 — CX / Process

Customer Support	chat_support	Chat support	**16**	5 — CX / Process
Customer Support	support_unhelp	Support unhelpful	**16**	5 — CX / Process
Customer Support	unhelp	Unhelpful	**16**	5 — CX / Process